AI Day — GitHub Agentic Workflows with Claude Code

Automatically triage issues and open PRs — no human in the loop.

What We Built

A GitHub Actions workflow that runs **Claude Code** on every new issue:

1. **Classifies** the issue — bug, enhancement, question, or needs-more-info
2. **Investigates** — reads relevant source code, traces execution paths
3. **Implements** — opens a `fix:` or `feat:` PR if confident
4. **Summarises** — always posts a triage comment with classification + outcome

Target codebase: a minimal **Next.js todo app** with a seeded off-by-one bug.

GitHub Agentic Workflows (gh–aw)

A GitHub CLI extension that compiles **Markdown workflow files** into standard Actions YAML.

```
.github/workflows/triage-and-implement.md        ← you edit this
.github/workflows/triage-and-implement.lock.yml  ← compiled, don't touch
```

The `.md` file has two parts:

- **YAML front matter** — trigger, engine, permissions, safe-outputs, tools
- **Markdown body** — the natural language prompt driving agent behavior

```
gh aw compile    # compile .md → .lock.yml (run after every edit)
gh aw validate   # validate without writing files
gh aw list       # show workflow status
```

Workflow Front Matter

```
on:
  issues:
    types: [opened]
engine: claude              # uses ANTHROPIC_API_KEY
permissions:
  contents: read
  issues: read
  pull-requests: read
safe-outputs:               # write actions, batched at end of run
  create-pull-request:
  add-labels:
  add-comment:
tools:
  github:
    toolsets: [default]
```

Key Concept: Safe Outputs

All **write actions are batched** and executed after the agent finishes — there is no intermediate output during a run.

- ✅ PRs, labels, and comments all appear together at the end
- ❌ No "I'm investigating..." acknowledgment comment mid-run
- ❌ Can't split acknowledgment and action into two `add-comment` calls

> **Implication:** accept the delay, or split into separate workflows chained via events — but chaining has its own gotchas (see slide 9).

Engine: Claude Code

```
engine: claude   # requires ANTHROPIC_API_KEY secret
```

- Full agentic loop: reads files, reasons, writes code, opens PRs
- Default engine (`copilot`) is lighter but less capable for code generation
- Claude Code is well worth it for **implementation tasks**

> **Lesson:** Use Copilot for pure classification/triage if cost matters. Use Claude Code when you want it to actually write code.

What the Prompt Does

```
Step 1: Classify  →  apply label
         ↓ (bug or enhancement)
Step 2: Gather Context  →  read relevant source files
         ↓
Step 3: Investigate & Implement
   If bug         →  find root cause  →  open fix: PR
   If enhancement →  assess feasibility  →  open feat: PR
         ↓ (always)
Step 4: Post triage summary comment
```

Only hard guards: **high confidence required**, no breaking changes.

The Seeded Bug

```
// lib/todos.ts
export function countRemaining(todos: Todo[]): number {
  let count = 0;
  for (let i = 1; i < todos.length; i++) {  // ← starts at 1, skips first todo
    if (!todos[i].completed) count++;
  }
  return count;
}
```

Issue filed: *"Item count in footer is always one less than expected"*

→ Agent finds the off-by-one, changes `i = 1` to `i = 0`, opens a PR. ✅

What Didn't Work: Two-Workflow Chaining

We tried splitting triage and fix into separate workflows chained via labels:

```
triage.md  →  adds label: triage:fix-queued
fix.md     →  triggers on: issues: labeled: triage:fix-queued
```

Problem 1: `GITHUB_TOKEN` (bot token) does not fire `issues: labeled` events that trigger other workflows — a GitHub security restriction.

Problem 2: Setting `GH_AW_GITHUB_TOKEN` to a real PAT fixed triggering, but caused **two fix workflows** to fire (one per label added by triage).

> **Decision:** Revert to single workflow. Simpler, and safe-outputs batching isn't a real limitation in practice.

Lessons Learned

#	Lesson
1	`safe-outputs` values must be **empty or `max:`**, not `true`
2	`tools` is an **object**, not an array
3	Strict mode forbids write permissions — use `safe-outputs` instead
4	All outputs are **batched** — no mid-run feedback in a single workflow
5	`GITHUB_TOKEN` cannot trigger other workflows — need a PAT for chaining
6	Always run `gh aw compile` after edits — catches schema errors early
7	Claude Code is powerful enough to implement enhancements, not just fix bugs

Repo Structure

```
.github/
  aw/actions-lock.json                # pinned action SHAs (like package-lock)
  workflows/
    triage-and-implement.md           # workflow source — edit this
    triage-and-implement.lock.yml     # compiled output — don't touch
app/
  page.tsx                            # Next.js todo UI
lib/
  todos.ts                            # utility logic (seeded bug lives here)
```

Deployed at: **https://ai-day-agentic-triage-fix.vercel.app**

PRs from the agent get **Vercel preview deployments** automatically.

Open an issue on **dcow/ai-day-agentic-triage-fix**

Watch Claude triage, investigate, and open a PR — all autonomously.